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## **Advancing Produced Water Injectivity: A Fit-For-Purpose Chemical Approach for Reuse and Disposal**

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### **Abstract**

Oilfield operations generate a significant volume of produced water (PW) containing large amounts of oil carryover, solids, heavy metals, and chemical additives. PW management is crucial for industry to demonstrate social responsibility and ensure the protection of ecosystems and human health. A key challenge in this process is maintaining injectivity into disposal (SWD) or injection wells to prevent excessive pressure buildup and formation damage.

In this work, PW samples collected from Salt-water Disposal (SWD) lines were characterized via particle size, turbidity and geochemical analysis. A laboratory workflow based on coreflood was developed to evaluate the injectivity of PW into the reservoir formation. Three chemically distinct additives—a standard iron sulfide dissolver (A), a novel non-corrosive dissolver (B), and a low molecular weight coagulant (C)—were initially tested. Injectivity index and permanent injectivity loss were measured for each treatment scenario. To validate laboratory findings under field-representative conditions, Millipore filtration tests were conducted on real-time produced water samples in the field, including evaluations of two additional formulations (D&E), specifically selected based on lab results for their enhanced ability to address oil carryover and manage suspended solids under variable field conditions.

Results show that in the absence of oil carryover, PW injection resulted in 80% injectivity loss mainly due to particle deposition and core face plugging. Treatments with chemicals A and B yielded a more than twofold improvement, reducing permanent injectivity loss to 27% and 29%, respectively. Coagulant C demonstrated exceptional performance when combined with a 25  $\mu\text{m}$  filtration step, reducing injectivity loss to just 2%. In contrast, the presence of oil carryover significantly impaired injectivity, leading to a 95% loss due to larger particle aggregates and increased oil wetting of the core surface. Chemical C, through the formation of insoluble flocs that enmeshed oil droplets and particles, achieved zero injectivity loss when coupled with filtration. Millipore tests further validated the superior performance of chemical treatment strategies, demonstrating improved filtration rates across all evaluated formulations.

Following the laboratory success, field trials were conducted in Permian Basin to validate the effectiveness of chemical treatment programs in enhancing both surface water quality and injectivity in saltwater disposal (SWD) wells. The chemical program, implemented across a multi-stage SWD line,

resulted in a 69% reduction in total suspended solids and an 88% reduction in oil carryover. Furthermore, a separate trial in a US South Delaware Basin disposal well demonstrated sustained injectivity improvements, as evidenced by consistently low Hall slope values over a 50-day treatment period. These results validate the chemical treatment strategy as a scalable, field-ready solution suitable for produced water reuse and disposal—particularly in water-stressed regions such as the Middle East and North Africa (MENA).

## Introduction

Produced water management has emerged as one of the most critical technical and economic challenges in oil and gas operations. With increasing global energy demand and expanding unconventional resource development, the volume of water requiring treatment, reuse, or disposal is growing rapidly. In the United States alone, the oilfield water management market is valued at over \$37.5 billion, reflecting both the scale of operations and the complexity of handling produced water effectively.

This complexity is especially evident in the Permian Basin, where advances in hydraulic fracturing have driven unprecedented water use. The total volume of water requiring management surged from 6.3 million barrels per day in 2017 to an estimated 18.9 million barrels per day in 2025 and is projected to reach 23.6 million barrels per day by 2030 ([The American Oil & Gas Reporter](#)). Meanwhile, the Middle East and North Africa (MENA) region faces a dual challenge: growing produced water volumes and severe water scarcity. According to the United Nations, 15 of the 22 Arab League countries fall below the water scarcity benchmark of 500 cubic meters per capita per year. Reflecting this urgency, the MENA produced water treatment market is projected to grow from USD 2.22 billion in 2024 to USD 3.38 billion by 2030, at a CAGR of 7.1% ([TechSci Research report](#)).

Maximizing produced water reuse or ensuring proper disposal under such demanding conditions requires technological innovation. One of the major hurdles is injectivity decline, which often results from formation damage caused by various physical, chemical, and biological mechanisms. These include:

- **Inorganic scaling**, such as silica, calcite, barium and calcium sulfate, iron sulfide, iron oxide, and salt precipitates that block pore throats; ([Jordan et al 2022](#); [Kermani et al 2023](#); [Cardenas et al 2024](#))
- **Organic deposition**, including paraffin and asphaltene accumulation, which can coat formation surfaces and reduce permeability ([Ly et al 1998](#); [Hurwitz et al 2015](#))
- **Microbial activity**, leading to biofilm formation or biomass plugging that restricts flow ([Hsi et al. 1994](#); [Ly et al 1998](#); [Jordan et al 2022](#)).
- **In-formation interactions**, such as emulsions, water blocks, wettability alteration, or clay swelling and migration, which disrupt natural flow paths and reduce injectivity ([Al-Shabibi et al 2022](#); [Al Moajil et al 2023](#))

Injectivity decline is rarely the result of a single mechanism; instead, it typically arises from the interplay of multiple, field-specific factors. Consequently, a combination of chemical and mechanical solutions is often required to manage water quality and maintain injectivity performance. Numerous field studies illustrate this integrated approach: [Hsi et al. \(1998\)](#) applied high-dosage biocide treatments and strong oxidizers to suppress microbial activity and restore permeability for produced water reinjection at Prudhoe Bay field in Alaska; [Jordan et al. \(2022\)](#) implemented a robust chemical program comprising mutual solvents, DTPA-based sulfate dissolvers, FeS dissolvers, biocides, surfactants, and scale inhibitors to remediate injectivity loss in a North Sea produced water reinjection (PWR) system; [Al-Shabibi et al. \(2022\)](#) recommended combining clay swelling inhibitors with advanced filtration systems to control suspended solids and prevent formation damage in an Omani oil field; [Ferrer and Bennett \(2022\)](#) developed a multifunctional chemical product designed to address residual hydrocarbon carryover, inorganic scaling, and FeS deposition in SWD systems, resulting in significant water quality improvements and enhanced disposal efficiency; [Shah et al. \(2024\)](#) introduced a biological oxidation method to reduce microbial load

and maintain hydrogen sulfide (H<sub>2</sub>S) concentrations below detection limits, thereby effectively eliminating sulfate-reducing bacteria (SRB) and preventing souring in saltwater disposal (SWD) systems. These examples underscore that successful water management is highly dependent on water chemistry, reservoir mineralogy, operational parameters, and even historical treatment practices. As a result, generic, one-size-fits-all chemical programs often fall short in complex, real-world environments.

This paper presents a fit-for-purpose chemical strategy aimed at improving injectivity for both water reuse and disposal. Drawing on technical insights from laboratory work and field operations in Permian basin, the approach can be specifically tailored for adaptation in water-stressed regions like MENA. By targeting specific damage mechanisms and customizing formulations to match local reservoir and water quality conditions, the strategy provides a scalable and sustainable path forward for produced water management.

## Experimental Methods

### Materials and Water Preparation

A produced water sample was collected from a saltwater disposal (SWD) line in the Permian Basin, with a total dissolved solids (TDS) concentration of 142,000 mg/L. A detailed geochemical analysis was performed, and the brine composition is summarized in [Table 1](#). The field sample contained no dispersed oil; therefore, a synthetic reserve emulsion was created to simulate oil carryover. This was achieved using an IKA homogenizer operating at 20,000 rpm for 5 minutes, yielding an emulsion with 200 ppm oil content in the water. The properties of the crude oil used for spiking as oil carryover are listed in [Table 2](#). Accordingly, water samples were prepared in two configurations for testing: (1) as received (no oil), and (2) spiked with 200 ppm oil simulating as carryover. Three chemical treatments—designated as Chemicals A, B, and C—were initially evaluated for their ability to reduce turbidity and improve injectivity with detailed formulations summarized in [Table 3](#). Each chemical was dosed at their respective treatment concentrations and mixed into the water using a magnetic stirrer at 300 rpm for 30 seconds to ensure adequate interaction. Chemicals A and B were tested without filtration. Chemical C, which functions by forming precipitates, was evaluated both with and without a filtration step to assess the impact of solids removal on performance. Injectivity was assessed using coreflood experiments conducted on San Andres outcrop cores (3 inches in length and 1.5 inches in diameter), obtained from Kocurek Industries. The porosity and permeability characteristics of the cores are presented in [Table 4](#). To further validate chemical treatment performance under field-representative conditions, Millipore filtration tests were conducted including evaluations of two additional formulations, D&E.

Table 1—Brine composition of the produced water sample used in this study

| Key cations, mg/L                     |        |
|---------------------------------------|--------|
| Barium (Ba)                           | 1.23   |
| Calcium (Ca)                          | 4260   |
| Iron (Fe)                             | 29.5   |
| Magnesium (Mg)                        | 899    |
| Manganese (Mn)                        | 1.35   |
| Sodium (Na)                           | 50200  |
| Silicon (Si)                          | 10.6   |
| Boron (B)                             | 45.6   |
| Strontium (Sr)                        | 382    |
| Anions, mg/L                          |        |
| Chloride (Cl)                         | 83200  |
| Bromide (Br)                          | 640    |
| Sulfate (SO <sub>4</sub> )            | 920    |
| Others                                |        |
| Specific gravity                      | 1.0961 |
| TDS, mg/L                             | 142000 |
| pH                                    | 7.1    |
| Bicarbonate (HCO <sub>3</sub> ), mg/L | 240    |

Table 2—Crude oil properties

| Property                        | Value |
|---------------------------------|-------|
| API Gravity                     | 22    |
| Wax Appearance Temperature (°C) | 13    |
| Wax Content ( $\geq$ C16, wt%)  | 1.8   |
| Pour Point (°C)                 | -29   |
| SARA Compositions               | wt%   |
| Saturates                       | 28.13 |
| Aromatics                       | 57.68 |
| Resins                          | 11.77 |
| Asphaltenes                     | 2.42  |

**Table 3—Chemical formulations used to treat the produced water for injectivity improvement**

| Water Treatment Solutions | Description                              |
|---------------------------|------------------------------------------|
| <b>A</b>                  | Industry standard iron sulfide dissolver |
| <b>B</b>                  | New non-corrosive iron sulfide dissolver |
| <b>C</b>                  | Low MW coagulant                         |
| <b>D</b>                  | A surfactant-based injectivity improver  |
| <b>E</b>                  | Surfactant based dispersant              |

**Table 4—properties of San Andres outcrop cores along with treatment solutions tested in CF**

| Core ID | Porosity (%) | Brine Perm (mD) | Injected Fluid                    | Treatment Fluid                               |
|---------|--------------|-----------------|-----------------------------------|-----------------------------------------------|
| #1      | 18.4         | 17              | Produced water                    | NA                                            |
| #2      | 18.9         | 19              | Produced water                    | 20-25 $\mu\text{m}$ filtration                |
| #3      | 19.1         | 16              | Produced water                    | 100 ppm A                                     |
| #4      | 19.5         | 19              | Produced water                    | 100 ppm B                                     |
| #5      | 19.9         | 15              | Produced water                    | 100 ppm C                                     |
| #6      | 18.3         | 15              | Produced water                    | 100 ppm C with 20-25 $\mu\text{m}$ filtration |
| #7      | 19.9         | 21              | Produced water +200 ppm Crude oil | NA                                            |
| #8      | 19.2         | 15              | Produced water +200 ppm Crude oil | 100 ppm A                                     |
| #9      | 19.2         | 14              | Produced water +200 ppm Crude oil | 100 ppm B                                     |
| #10     | 19.2         | 15              | Produced water +200 ppm Crude oil | 100 ppm C with 20-25 $\mu\text{m}$ filtration |

## Particle Sizer

Particle size distribution (PSD) of the water samples was analyzed using a Horiba Partica LA-950V2 laser diffraction particle size analyzer. This instrument uses laser scattering technology to measure particles ranging from 0.01  $\mu\text{m}$  to 3000  $\mu\text{m}$  in diameter, providing high-resolution size data across a wide dynamic range.

Prior to analysis, samples were gently agitated to ensure homogeneity. Approximately 5 mL of sample was added to the instrument's dispersion unit, which was filled with deionized water as the dispersant. The built-in stirrer was operated at 200 rpm to prevent particle settling during measurement. Refractive index values for water and typical solid contaminants were input into the software for accurate calculation. Each sample was measured in triplicate to ensure reproducibility, and the results were reported as volume-based particle size distributions.

## Turbidity

Turbidity was measured to evaluate water quality before and after chemical treatment. For each test, 20 mL of water—treated or untreated—was transferred into a clean scintillation vial. The vial was then placed in a Turbiscan analyzer, which scanned the sample every 30 seconds for a total duration of 10 minutes.

The average light transmittance across the full height of the sample column was recorded and used as an indicator of water quality.

### Coreflood Injectivity

A coreflood (CF) setup was constructed to evaluate the effectiveness of chemical treatments in improving injectivity. The system consisted of a core holder, an Isco pump, Quizix pump, pressure transducer, and an effluent collection container, as illustrated in Figure 1.

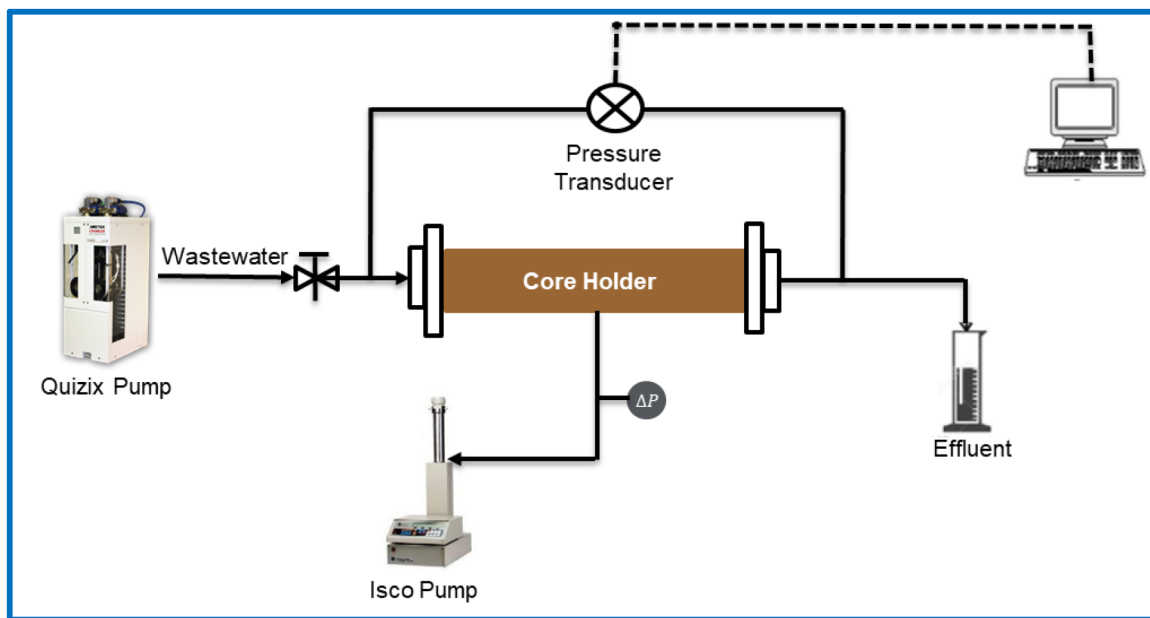


Figure 1—Coreflood setup used for produced water injectivity tests

Core samples (described in Table 4) were first vacuum-saturated with synthetic brine (SB) and placed in the core holder. Initial permeability was measured using SB to establish a baseline pressure drop. Following baseline measurement, field produced water, either untreated or treated with chemical solutions, was injected for a total of 100 pore volumes (PVs). Subsequently, the core was flushed with synthetic brine to assess residual or permanent injectivity loss. Injectivity performance was quantified using the injectivity index ( $\Pi$ ), defined by the following equation (Civan, 2023):

$$\Pi = Q / \Delta P$$

$$\text{Injectivity Loss (\%)} = \left(1 - \frac{\Pi}{\Pi_0}\right) \times 100$$

Where:

- $\Pi$  is the injectivity index after water injection (with or without treatment)
- $\Pi_0$  is the initial injectivity index measured with synthetic brine
- $Q$  is the flow rate
- $\Delta P$  is the pressure drop across the core

### Millipore Testing

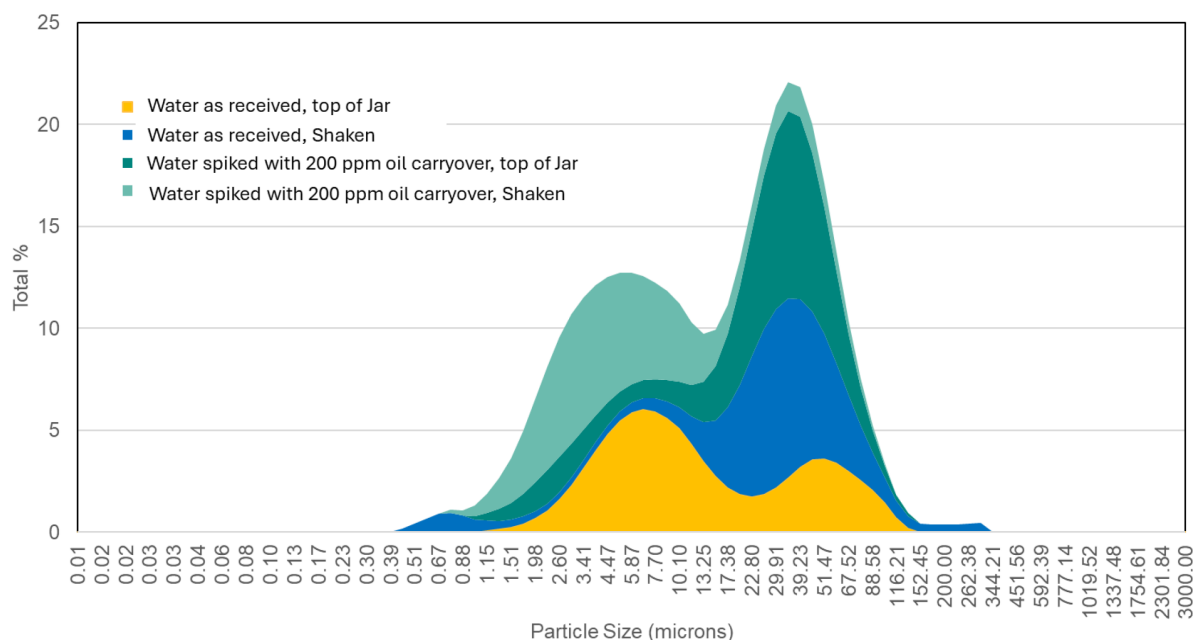
The Millipore test was performed to evaluate total suspended solids (TSS) and filtration performance of water samples. The procedure involved the following steps:

- **Filter Preparation:** A clean, dry 0.45  $\mu\text{m}$  Millipore membrane filter was weighed using an analytical balance and recorded as the initial dry weight
- **Filter Wetting:** The membrane filter was wetted with deionized water to ensure proper adhesion and placed securely in the filter holder
- **Sample Loading and Pressurization:** The water sample was poured into a stainless-steel pressure vessel connected to the filter assembly. The vessel was sealed and pressurized to 20 psi using compressed air or nitrogen
- **Filtration and Timing:** The outlet valve was opened to initiate filtration. Filtrate flow was collected into a graduated cylinder, and time was recorded at every 100 mL interval. Filtration was continued until either 1 liter of water had passed through or 5 minutes had elapsed (whichever occurred first)
- **Post-Filtration Handling:** After filtration, the Millipore filter was carefully removed, dried in an oven, and reweighed to determine the final dry weight. The total suspended solids (TSS) were calculated as the difference between the final and initial weights of the filter, normalized to the volume of water filtered (mg/L)

## Laboratory Results and Discussion

Particle size distribution analysis was conducted to evaluate the impact of oil carryover and agitation on suspended solids in produced water samples. Four conditions were tested: water as received (top of jar), water as received after agitation (shaken), water spiked with 200 ppm oil carryover (top of jar), and water spiked with 200 ppm oil carryover followed by agitation (shaken), as shown in [Figure 2](#). The baseline sample (top of jar, as received) exhibited a bimodal PSD with peaks around 10–20  $\mu\text{m}$  and a broader distribution from 30 to 70  $\mu\text{m}$ , suggesting that under static conditions, heavier particles had settled, and finer particles remained suspended. After agitation, the PSD shifted toward higher particle counts in the 20–60  $\mu\text{m}$  range, indicating resuspension of settled particles due to mechanical disturbance. When oil carryover was introduced without agitation, the PSD showed increased particle counts, particularly in the 10–50  $\mu\text{m}$  range, suggesting that oil may help stabilize or disperse solids, preventing settling. The combination of oil and agitation produced the highest total suspended solids across nearly the entire size range, especially from 10–60  $\mu\text{m}$ , indicating a synergistic effect where oil likely facilitated the formation or stabilization of emulsified or agglomerated particles. These results demonstrate that both agitation and oil carryover significantly affect the particle size distribution and solids concentration in produced water, with important implications for injectivity impairment due to the unfavorable interaction between particles, oil carryover and reservoir rock. The enhanced suspension of solid particles in the presence of oil carryover may present further challenges for separation efficiency and impact water injectivity performance, underscoring the need for targeted strategies to manage these effects in field operations. Supporting these observations, analysis of filtrate from a SWD Millipore field test confirmed that 14% of the solids were iron compounds. Iron particles tend to interact with crude oil and become strongly oil-wet, which is consistent with the behavior observed in these PSD results.





**Figure 2—Particle size distribution (PSD) of produced water under four conditions: as received (top of jar), as received with agitation (shaken), spiked with 200 ppm oil carryover (top of jar), and spiked with 200 ppm oil carryover followed by agitation (shaken).**

Turbiscan analysis was conducted to assess water clarity under different chemical treatments as well as the impact of oil carryover. Transmission percentage, an inverse indicator of suspended solids and emulsion stability, was used to evaluate water quality. As shown in Figure 3, the "water as received" sample exhibited high transmission (~88%), indicating low turbidity and minimal suspended solids. However, when spiked with 200 ppm oil carryover, the transmission dropped significantly (~36%), confirming that oil promotes dispersion or stabilization of fine solids, leading to reduced clarity. Upon treatment with 100 ppm of chemicals A and B in the presence of oil, transmission remained low, with 100 ppm A showing a drastic decrease (<5%), suggesting poor performance in mitigating the oil-induced turbidity. In contrast, 100 ppm C combined with 20–25  $\mu\text{m}$  filtration showed substantial improvement, with transmission increasing to ~80%, indicating effective removal or stabilization of dispersed solids and improved water clarity. Supporting this, geochemical analysis (Figure 4) confirmed the complete removal of iron species, with concentrations reduced from 29.5 to 0 mg/L in the treated water, while levels of other ions—such as calcium, magnesium, and sodium—remained largely unchanged. These results highlight the importance of selecting appropriate chemical treatments and mechanical filtration to manage solids behavior in oily produced water. While this result is promising, the field applicability of chemical C may be limited by the lack of appropriate filtration infrastructure and solid management protocols required to handle separated solids. Therefore, the preferred strategy remains to keep particles dispersed in a stable, manageable state to prevent deposition.



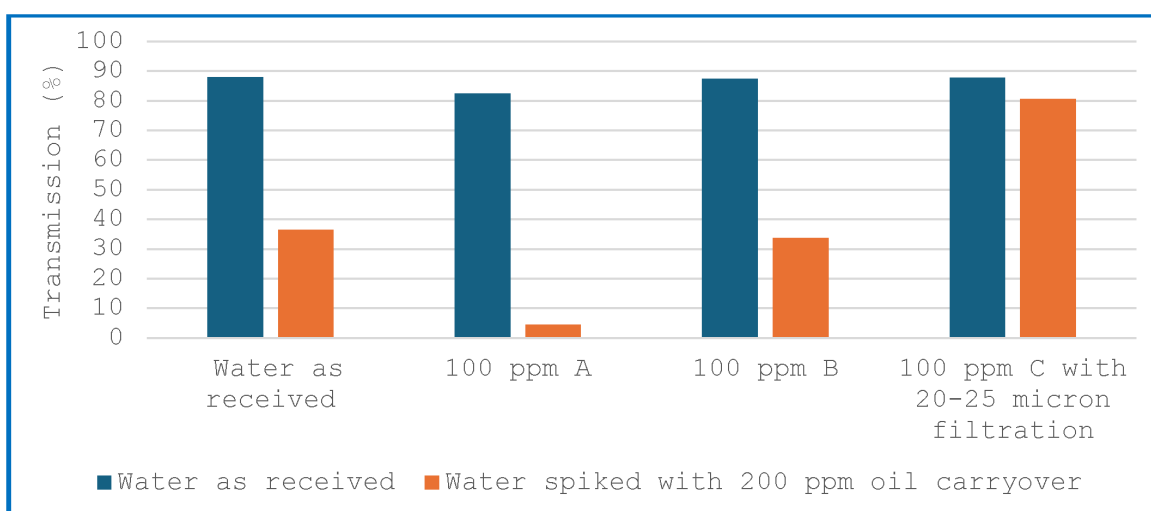


Figure 3—Transmission (%) measurements from Turbiscan analysis to evaluate water clarity under different conditions.

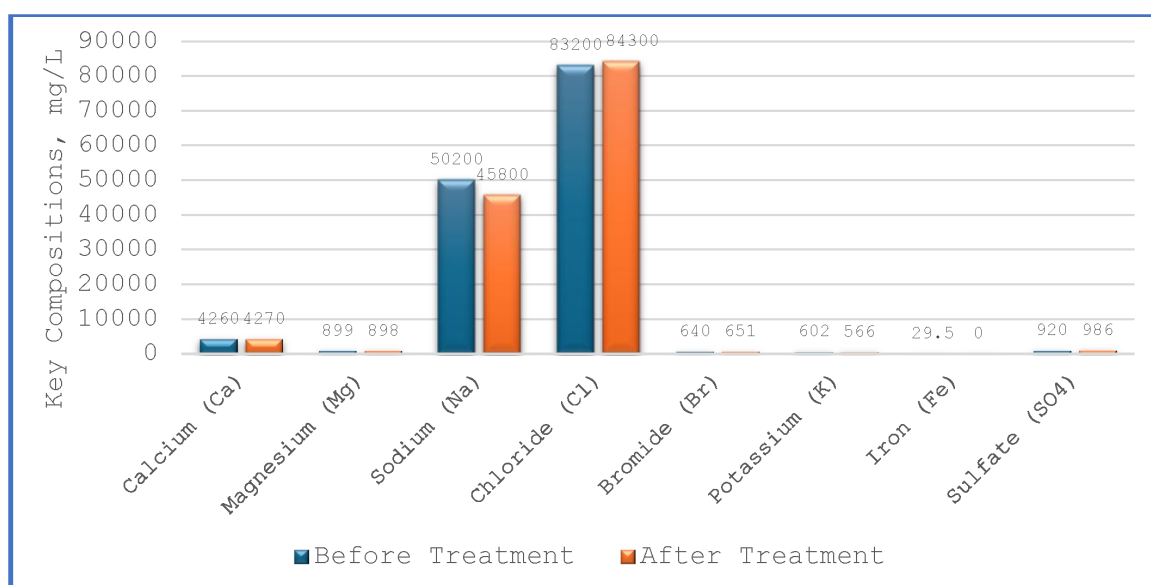


Figure 4—Brine composition comparison for produced water treated by 100 ppm C and 20-25 mm filtration

To further evaluate the performance of these chemicals, injectivity performance was evaluated by measuring pressure buildup during produced water injection and recovery after a flush step under various treatment conditions, with and without crude oil carryover using coreflood. The injectivity loss after injection and flush was used as a key indicator of treatment effectiveness. As summarized in Table 5, in untreated produced water, both unfiltered and filtered samples showed high injectivity loss after injection (79.81% and 79.95%, respectively), with some improvement after the flush (74.71% and 52.35%). This indicates that filtration alone without chemical treatment is insufficient to mitigate formation plugging. Chemical treatments with 100 ppm of A, B, or C showed improved performance in produced water without oil. After flush, injectivity loss decreased to 26.86% for A, 28.99% for B, and 51.93% for C. However, combining 100 ppm of chemical C with 20–25  $\mu$ m filtration drastically reduced injectivity loss to just 5.70% after flush, highlighting a synergistic benefit between chemical and mechanical treatments in stabilizing or removing suspended solids. In the presence of 200 ppm oil carryover, untreated water led to severe plugging with 95.32% injectivity loss after injection and 89.64% remaining after flush. The addition of chemical A worsened this outcome, resulting in the highest pressure buildup and 92.47% loss after flush,

suggesting that A may have exacerbated emulsion or solids stabilization under oily conditions. Chemical B performed better in the presence of the oil carryover, reducing post-flush loss to 51.16%. These results align with Turbiscan water quality measurements and reinforce the importance of selecting chemical treatments based on the presence of oil carryover. Chemical A outperformed B in less oily water conditions, while Chemical B proved more effective in managing oily water. Notably, the combination of 100 ppm C with 20–25  $\mu\text{m}$  filtration again delivered superior performance, achieving 0% injectivity loss, indicating effective mitigation of oil and particle-induced damage.

**Table 5—Injectivity performance under various chemical and mechanical treatments for produced water with and without oil carryover. Initial pressure, post-injection pressure, and post-flush pressure were recorded to evaluate treatment effectiveness. Injectivity loss (%) was calculated after produced water injection and subsequent flush.**

| Samples                      | Treatment                     | Initial Pressure, psi | Post-Injection Pressure, psi | Post-Flush Pressure, psi | Injectivity loss after injection, % | Injectivity loss after flush, % |
|------------------------------|-------------------------------|-----------------------|------------------------------|--------------------------|-------------------------------------|---------------------------------|
| Water as received            | None                          | 8.60                  | 42.60                        | 34.00                    | 79.81%                              | 74.71%                          |
| Water as received            | 20-25 mm filtration           | 9.11                  | 45.44                        | 19.12                    | 79.95%                              | 52.35%                          |
| Water as received            | 100 ppm A                     | 8.85                  | 18.60                        | 12.10                    | 52.42%                              | 26.86%                          |
| Water as received            | 100 ppm B                     | 7.3                   | 17.11                        | 10.28                    | 57.33%                              | 28.99%                          |
| Water as received            | 100 ppm C                     | 9.5                   | 21.12                        | 14.99                    | 55.02%                              | 51.93%                          |
| Water as received            | 100 ppm C+20-25 mm filtration | 8.94                  | 9.12                         | 9.48                     | 1.97%                               | 5.70%                           |
| Water+ 200 ppm oil carryover | None                          | 7.05                  | 150.46                       | 67.99                    | 95.32%                              | 89.64%                          |
| Water+ 200 ppm oil carryover | 100 ppm A                     | 11.24                 | 338.60                       | 149.24                   | 96.68%                              | 92.47%                          |
| Water+ 200 ppm oil carryover | 100 ppm B                     | 11.6                  | 48.98                        | 23.75                    | 76.32%                              | 51.16%                          |
| Water+ 200 ppm oil carryover | 100 ppm C+20-25 mm filtration | 10.94                 | 10.84                        | 10.26                    | 0.00%                               | 0.00%                           |

Based on the earlier PSD, turbidity, and injectivity studies, it is evident that suspended solids and oil carryover can significantly impact water quality and, consequently, the performance of chemical treatments. Additionally, produced water quality is known to vary under field conditions. To ensure that chemical selection reflects field-relevant performance, Millipore filtration tests were conducted in a field laboratory using real-time produced water samples. These tests provide a practical indication of how well solids and oil carryover can be managed under realistic conditions. In this series of Millipore tests, two additional chemistries—Chemical D and E—were evaluated alongside previously tested treatments (Chemical A and B). Chemical C, despite its strong laboratory performance, was excluded due to the lack of suitable filtration infrastructure at the field site. The newly introduced chemistries (D and E) were specifically designed to further improve solid wetting, facilitate separation of solids from oil, minimize particle aggregation, and maintain a stable dispersion of solids in solution—key goals for reducing deposition and improving water

injectivity. As shown in Figure 5, untreated produced water exhibited the lowest throughput, indicating high plugging potential due to poor water clarity and particle accumulation. Among the treated samples, both Chemical A and B showed improvement over the untreated baseline but remained limited in performance. Notably, Chemical D and E outperformed all other treatments, yielding the highest filtration rates over time, ability to manage solids under field conditions, likely due to their enhanced ability to separate oil-coated particles and maintain dispersion. These findings reinforce the importance of testing chemicals using representative produced water under field-relevant conditions. Laboratory screening provides valuable insight, but field testing such as the Millipore test is essential to validate performance, especially in systems with variable oil and solids content.

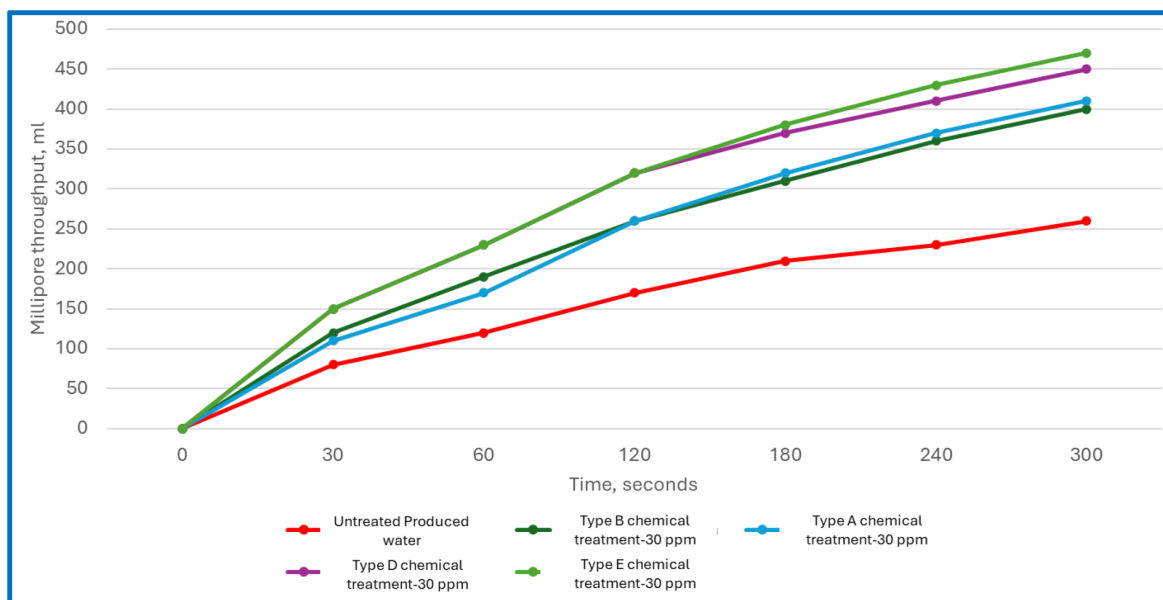


Figure 5—Millipore filtration throughput over time for untreated produced water and produced water treated with various chemical additives

## Field Trial Results

Following laboratory studies, several field trials were conducted in the Permian Basin to validate the effectiveness of chemical treatment programs in improving both surface water quality and injectivity in saltwater disposal (SWD) wells.

In one field example, a multi-stage chemical treatment process was implemented across a central gathering facility (CGF), pipeline, and SWD inlet. Three tailored chemistries (A, B, E) were applied at each location to target total solids and oil carryover (Figure 6). This program began in July 2024. Post-treatment monitoring demonstrated substantial surface-level water quality improvements, with total solids reduced by an average of 69% and oil carryover decreased by 88%. These improvements are illustrated in Figures 7 and 8.

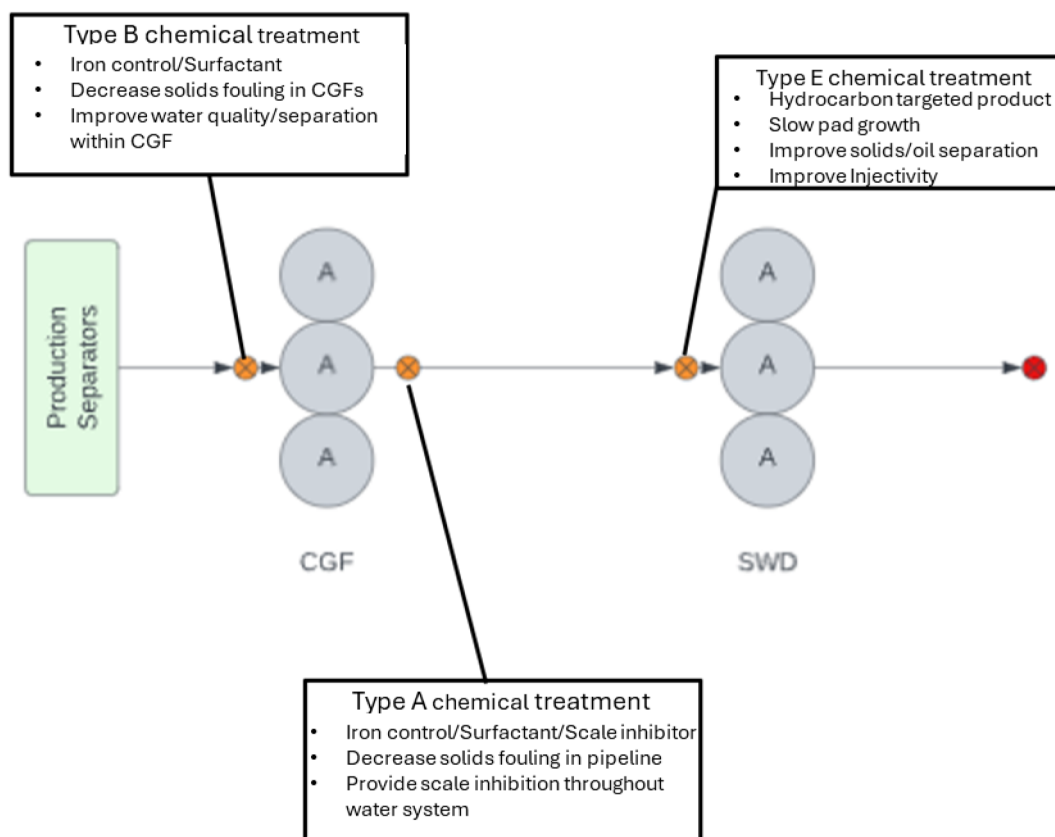


Figure 6—Multi-stage chemical treatment process deployed at the CGF, pipeline, and SWD inlet

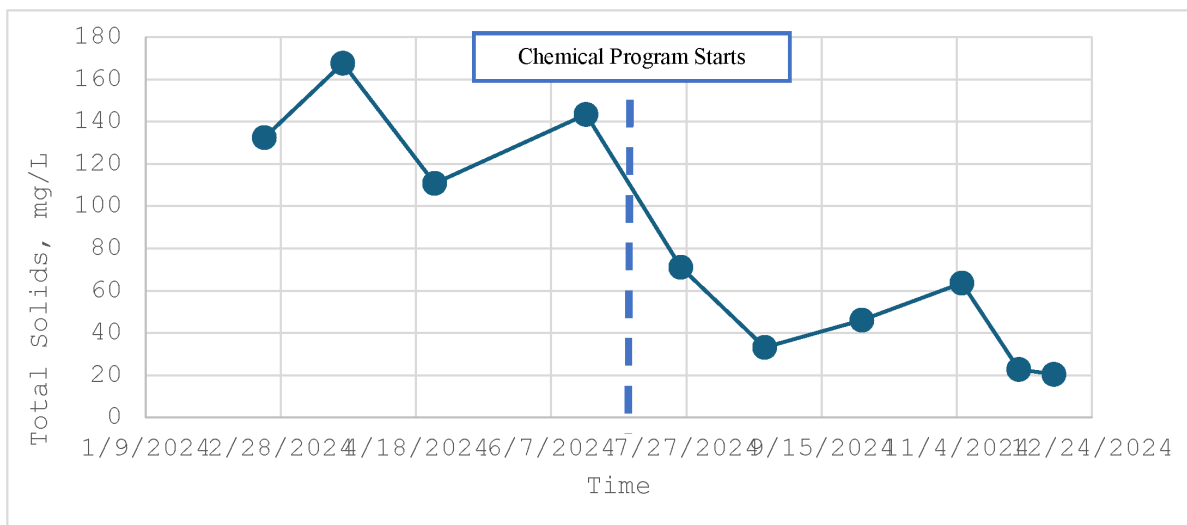


Figure 7—Total solid before and after the chemical treatment

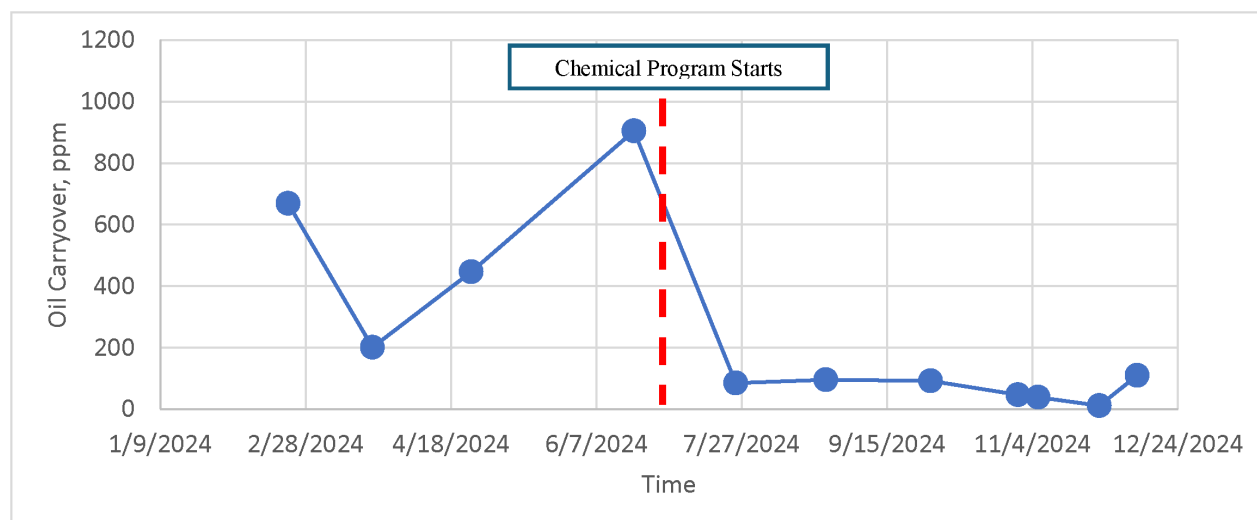


Figure 8—Oil carryover before and after the chemical treatment

In addition to surface-level treatment, a separate field trial was conducted on a disposal well in the South Delaware Basin to assess downhole injectivity. In this trial, an iron control product was dosed at the surface facility, followed by a surfactant-based injectivity improver (Chemical D) to the injection stream. This surfactant was designed to alter the wettability of suspended solids, keeping them dispersed in the water phase and preventing accumulation or plugging. The injectivity improvement was assessed by the Hall slope type curve using the following equation (Hall H.N 1963):

$$\text{Hall Slope} = \frac{WHP}{Q}$$

Where WHP is the wellhead pressure and  $Q$  is the produced water injection rate.

Figure 9 presents the Hall plot results. Chemical treatment began in September 2023, with an increased dosage implemented in October and continued until February 1, 2024. For more than 50 days during the treatment period, a consistently low Hall slope was observed, indicating improved injectivity. However, when the chemical program was halted on February 11, 2024, the Hall slope began to rise, signaling a decline in injectivity (Figure 10). These observations confirm the effectiveness of the chemical program in enhancing and sustaining injectivity performance in the field.

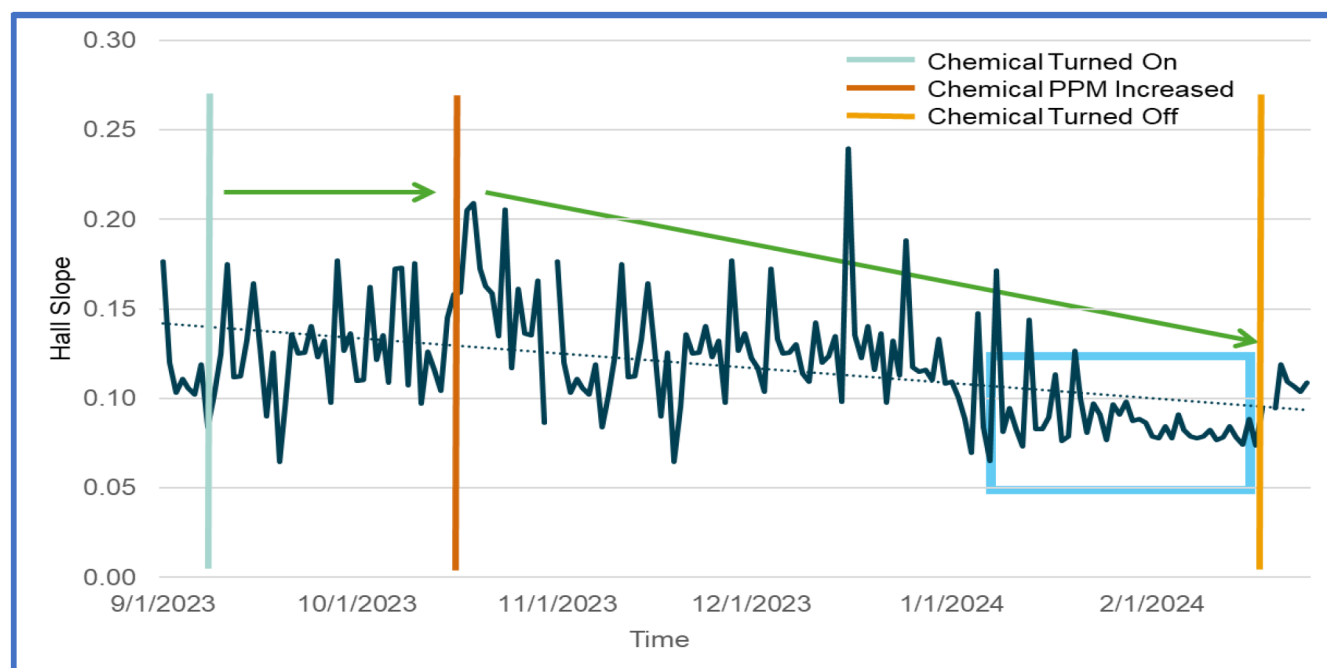


Figure 9—Changes in injectivity observed in a Hall plot following chemical treatment of a disposal well in the South Delaware Basin.

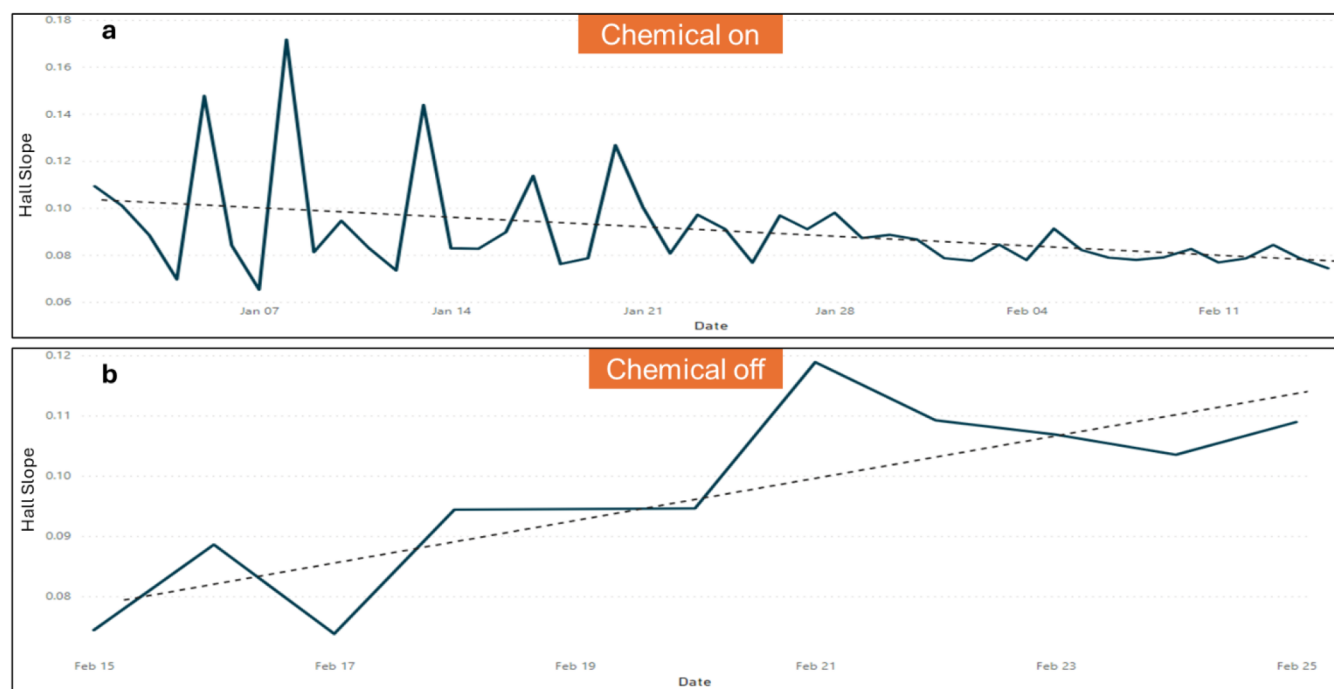


Figure 10—Increase in Hall slope indicating injectivity decline following discontinuation of chemical treatment: (a) chemical treatment on; (b) chemical treatment off.

## Conclusions

This study presents a chemical treatment strategy designed to improve produced water (PW) injectivity for both reuse and disposal applications. Laboratory evaluations demonstrated that suspended solids and oil carryover are primary contributors to injectivity impairment. In the presence of oil, untreated PW resulted in injectivity losses as high as 95%. However, the application of tailored chemical solutions effectively

reduced formation damage and mitigated injectivity loss, with chemical selection dependent on the levels of solids and oil carryover present in the water.

To ensure field-relevant performance, Millipore filtration tests were conducted using real-time PW samples, which validated the effectiveness of improved chemical formulations under operational conditions. Field deployment of the treatment program across a multi-stage saltwater disposal (SWD) system achieved a 69% reduction in total solids and an 88% reduction in oil carryover on surface. Downhole improvements in the disposal well were confirmed via Hall plot analysis, which showed sustained lower injectivity over a 50-day treatment period. A sharp rise in Hall slope following the discontinuation of chemical injection further confirmed the program's impact.

Overall, this work highlights the importance of aligning chemical treatment strategies with site-specific water chemistry and infrastructure limitations. The demonstrated approach offers a scalable, effective solution for PW management and injectivity enhancement, with strong potential for deployment in water-stressed regions such as the Middle East and North Africa (MENA), where sustainable water handling is a growing priority.

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